РЕДОВЕ

I РЕДОВЕ С НЕОТРИЦАТЕЛНИ ЧЛЕНОВЕ

① $\sum \left(n \operatorname{arctg} \frac{1}{n^2}\right)^{n^5}$

② $\sum n^{n^2/2} \left(\operatorname{arctg} \frac{1}{\sqrt{n}}\right)^{n^2}$

③ $\sum \frac{1}{(\ln n)^{\ln \ln n}}$

④ $\sum \frac{1}{\ln^2 n} \left(e - \left(1+\frac{1}{n^2}\right)^{n^2}\right)^{\alpha}$

⑤ $\sum \left(\operatorname{arctg} \frac{1}{n^3}\right)^n (3n)!$

⑥ $\sum \left(1 - n \sin \frac{1}{n}\right)^{\alpha}$

⑦ $\sum \left(\cos \frac{1}{\sqrt{n}} - \frac{\sqrt{n^2-n}}{n}\right)^{\alpha}$

⑧ $\sum \left(\frac{1}{\sqrt{n}} - \sqrt{\ln \frac{n+1}{n}}\right)^{\alpha}$

⑨ $\sum \left\{\left(e^{1/n} - \sin \frac{1}{n}\right)^{n^{\alpha}} - 1\right\}$

⑩ $\sum \frac{\ln (e^n + n^2)}{n^2 \ln^2 (n+1)}$

⑪ $\sum \left(1 - \frac{1}{n \sqrt{\ln (n+1)}}\right)$

⑫ $\sum \frac{1}{\ln^2 n} \left(e - \left(1+\frac{1}{n^2}\right)^{n^2}\right)^{\alpha}$

⑬ $\sum \frac{\operatorname{arctg} \left(e - \left(1+\frac{1}{n}\right)^n\right)}{\ln^{\alpha} (1+n^2)}$

⑭ $\sum \frac{n}{(\ln n)^{n/2}}$

II РЕДОВЕ СЪС ЗНАКОПРОМЕНЛИВИ ЧЛЕНОВЕ

⑮ $\sum \frac{\cos n}{n^{2/3} + (-1)^n}$

⑯ $\sum \left(\sqrt[n]{n} - 1\right) \cos n$

⑰ $\sum \left(e^{\cos n/\sqrt{n}} - \cos \frac{1}{n}\right)$

⑱ $\sum \frac{1}{\sqrt[4]{n}} \cos \frac{\pi (n^2+n)}{n}$

⑲ $\sum \frac{\sin (n + 1/n)}{\ln n \ln (n+1)}$

⑳ $\sum (-1)^n \frac{\sqrt{n+1}}{\sqrt{n}+2}$

㉑ $\sum \cos \left(\pi n + \frac{\pi}{6}\right) \ln \left(1 + \frac{2}{n}\right)$

㉒ $\sum \frac{(-1)^n}{n^{\alpha} - (-1)^n}$

㉓ $\sum \ln \left(1 + \frac{(-1)^{n+1}}{\sqrt{n}}\right)$

㉔ $\sum \left[\frac{\sin n}{\sqrt[3]{n^2}} - \sin \left(\frac{\sin n}{\sqrt[3]{n^2}}\right)\right]$

㉕ $\sum (-1)^n \left(\operatorname{arctg} \frac{1}{n} - \arcsin \frac{1}{n}\right)$